Automated Experiment Report

Run folder: run_20260419_203159

Prepared on 2026-04-19 18:12:52. This report was created automatically from saved run artifacts using deterministic Python rules only.

Model family Direct Hurdle Tabular	**Horizons** 24, 48
Platform Linux-6.6.113+-x86_64-with-glibc2.35	**XGBoost device** cuda
Classifier configs 2	**Regressor configs** 2
Pairs per horizon 4	**Total fold fits** 8

Executive summary

- The run archived 142 tracked artifacts.
- Best 24h pair: h24__clf_xgbc01__reg_xgbr01.
- Best 48h pair: h48__clf_xgbc01__reg_xgbr01.
- For horizon 24h, the winning pair was Xgb Classifier + Xgb Regressor. Average county RMSE expected was 97.2158; Improved by 3.0895 (3.08%). Stage-1 ROC AUC was 0.8999, indicating strong discrimination.
- For horizon 48h, the winning pair was Xgb Classifier + Xgb Regressor. Average county RMSE expected was 73.2486; Improved by 2.1075 (2.80%). Stage-1 ROC AUC was 0.8733, indicating strong discrimination.
- Submission-style CSVs were exported from: Best validation candidate.
- Demo-only evaluation files are present. They are useful for sanity checks only and are not the hidden challenge evaluation.

Configuration and modeling decisions

- Architecture: the run uses a two-stage direct hurdle pipeline ('direct_hurdle_tabular'). Stage 1 predicts whether outage is non-zero; Stage 2 predicts outage size conditional on positive outcomes only.
- Combination rule: the final expected outage equals stage-1 non-zero probability times the stage-2 positive prediction.
- Target transform: Stage 2 uses log1p on positive outages before fitting the regressor.
- Forecast horizons: 24h, 48h.
- Forecast-origin sampling: every eligible hour was used.
- Minimum training history before the first fold: 21 days (504 h).
- Maximum training history: capped at 28 days (672 h).

- Temporal validation design: manual aligned splits were specified - 24h train starts at 2023-06-01 00:00:00 and validates from origin 2023-06-11 23:00:00; 48h train starts at 2023-06-01 00:00:00 and validates from origin 2023-06-11 23:00:00.
- Target lag features: 24, 48 hours.
- Target rolling windows: 6 hours.
- Weather lag features: 24, 48 hours.
- Weather rolling windows: 6 hours.
- Weather screening: up to 10 weather variables were allowed; 10 were retained in this run.
- Weather outlier handling: enabled with 'winsorize_iqr', so extreme weather values were clipped rather than turned into missing values.
- Execution hardware: GPU acceleration was available; XGBoost used device 'cuda'.
- Prediction export policy: submission CSVs were generated from Best validation candidate.

Search space summary

component	families	config_count
Classifier	logistic_enet, xgb_classifier	2
Regressor	elasticnet_regression, xgb_regressor	2

Run inventory

The table below shows the top-level artifact folders detected in the run. The checkpoints folder is runtime infrastructure and may be intentionally absent from shared GitHub uploads.

subfolder	present	file_count	note
tables	Yes	33	
predictions	Yes	12	
models	Yes	40	
metrics	Yes	28	
plots	Yes	17	
checkpoints	Yes	10	Runtime-only resume artifacts; may be omitted from shared GitHub uploads.

Temporal evaluation design

The backtest table below maps the training and validation windows used for each horizon and fold.

horizon	fold_id	val_origin_idx	val_origin_timestamp	train_origin_start_idx	train_origin_start_timestamp	train_origin_max_idx	train_origin_max_timestamp	validation_window_start	validation_window_end
24	1	1,727	2023-06-11 23:00:00	1,464	2023-06-01	1,703	2023-06-10 23:00:00	2023-06-12	2023-06-12 23:00:00
48	1	1,727	2023-06-11 23:00:00	1,464	2023-06-01	1,679	2023-06-09 23:00:00	2023-06-12	2023-06-13 23:00:00

Feature frame and data overview

metric	value
rows_in_model_df	179,363
base_numeric_features	74

metric	value
categorical_features	1.0000
feature_warmup_hours	48
max_horizon	48
mean_outage	45.2483
median_outage	0.0000
std_outage	452.2684
max_outage	23,346
zero_rate	0.7047
nonzero_rate	0.2953

Selected weather features retained in this run: 10.

feature	abs_corr_with_outage	signed_corr_with_outage
gh_4	0.1299	0.1299
cape_1	0.0982	0.0982
pwat	0.0885	0.0885
sh2	0.0832	0.0832
mslma	0.0758	-0.0758
sdlwrf	0.0707	0.0707
mcc	0.0662	0.0662
d2m	0.0650	0.0650
cape	0.0641	0.0641
lcc	0.0554	0.0554

Candidate ranking by horizon

Rows are ranked within each horizon using average county RMSE expected first and overall RMSE expected second.

rank	horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage2_positive_rmse	pair_tag
1	24	xgb_classifier	xgb_regressor	97.2158	279.6099	3.0895	12.9494	0.8999	461.6680	h24__clf_xgbc01__reg_xgbr01
2	24	logistic_enet	xgb_regressor	97.4957	279.2281	2.8096	13.3312	0.8988	461.6680	h24__clf_logen01__reg_xgbr01
3	24	xgb_classifier	elasticnet_regression	98.1725	282.5314	2.1328	10.0278	0.8999	467.7318	h24__clf_xgbc01__reg_enet01
4	24	logistic_enet	elasticnet_regression	98.5449	282.3079	1.7605	10.2514	0.8988	467.7318	h24__clf_logen01__reg_enet01
1	48	xgb_classifier	xgb_regressor	73.2486	200.2759	2.1075	9.6489	0.8733	345.0510	h48__clf_xgbc01__reg_xgbr01

rank	horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage2_positive_rmse	pair_tag
2	48	logistic_enet	xgb_regressor	73.7777	199.8877	1.5784	10.0372	0.8763	345.0510	h48__clf_logen01__reg_xgbr01
3	48	xgb_classifier	elasticnet_regression	74.1479	202.8451	1.2082	7.0797	0.8733	351.1799	h48__clf_xgbc01__reg_enet01
4	48	logistic_enet	elasticnet_regression	74.9798	202.7862	0.3763	7.1387	0.8763	351.1799	h48__clf_logen01__reg_enet01

Winning pairs by horizon

horizon	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected	avg_county_gain_vs_zero	overall_gain_vs_zero	mean_stage1_roc_auc	mean_stage1_f1	mean_stage2_positive_rmse	pair_tag
24	xgb_classifier	xgb_regressor	97.2158	279.6099	3.0895	12.9494	0.8999	0.7143	461.6680	h24__clf_xgbc01__reg_xgbr01
48	xgb_classifier	xgb_regressor	73.2486	200.2759	2.1075	9.6489	0.8733	0.6493	345.0510	h48__clf_xgbc01__reg_xgbr01

Horizon 24h selected Xgb Classifier + Xgb Regressor. Average county RMSE expected was 97.2158 versus a zero baseline of 100.3053. Improved by 3.0895 (3.08%).

Horizon 48h selected Xgb Classifier + Xgb Regressor. Average county RMSE expected was 73.2486 versus a zero baseline of 75.3561. Improved by 2.1075 (2.80%).

Stage-specific summaries

Stage 1 classifiers

horizon	classifier_name	classifier_params	mean_stage1_roc_auc	mean_stage1_average_precision	mean_stage1_brier_score	mean_stage1_f1
24	xgb_classifier	{"n_estimators": 150, "learning_rate": 0.08, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	0.8999	0.8704	0.1247	0.7143
24	logistic_enet	{"C": 0.5, "l1_ratio": 0.15, "class_weight": "balanced", "max_iter": 250}	0.8988	0.8605	0.1209	0.7638

horizon	classifier_name	classifier_params	mean_stage1_roc_auc	mean_stage1_average_precision	mean_stage1_brier_score	mean_stage1_f1
48	logistic_enet	{"C": 0.5, "l1_ratio": 0.15, "class_weight": "balanced", "max_iter": 250}	0.8763	0.8266	0.1314	0.7267
48	xgb_classifier	{"n_estimators": 150, "learning_rate": 0.08, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	0.8733	0.8254	0.1312	0.6493

Stage 2 regressors

horizon	regressor_name	regressor_params	mean_stage2_positive_rmse	mean_stage2_positive_mae	mean_stage2_positive_rows_train
24	xgb_regressor	{"n_estimators": 200, "learning_rate": 0.06, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	461.6680	174.3335	140,227
24	elasticnet_regression	{"alpha": 0.0005, "l1_ratio": 0.1, "max_iter": 4000}	467.7318	179.1733	140,227
48	xgb_regressor	{"n_estimators": 200, "learning_rate": 0.06, "max_depth": 5, "subsample": 0.8, "colsample_bytree": 0.7, "min_child_weight": 6, "reg_alpha": 0.0, "reg_lambda": 1.5, "early_stopping_rounds": 15}	345.0510	107.9875	251,115
48	elasticnet_regression	{"alpha": 0.0005, "l1_ratio": 0.1, "max_iter": 4000}	351.1799	112.4882	251,115

Submission and refit artifacts

Submission model summary

horizon	prediction_model_source	pair_tag	fold_id	classifier_name	regressor_name	avg_county_rmse_expected	overall_rmse_expected
24	Best validation candidate	h24__clf_xgbc01__reg_xgbr01	1	xgb_classifier	xgb_regressor	97.2158	279.6099

horizon	prediction_model_source	pair_tag	fold_id	classifier_name	regressor_name	avg_county_rmse_expected	overall_rmse_expected
48	Best validation candidate	h48__clf_xgbc01__reg_xgbr01	1	xgb_classifier	xgb_regressor	73.2486	200.2759

Final refit summary

horizon	bundle_source	pair_tag	classifier_name	regressor_name	mean_avg_county_rmse_expected	mean_overall_rmse_expected
24	Final refit	h24__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	97.2158	279.6099
48	Final refit	h48__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	73.2486	200.2759

Prediction exports

file	rows	prediction_column	min	max	mean	std
two_stage_pred_24h.csv	1,992	pred	0.1410	113.7487	5.3191	15.9509
two_stage_pred_48h.csv	3,984	pred	0.1961	184.4337	6.4976	21.3125
final_refit_pred_24h.csv	1,992	pred	0.0425	243.1345	7.7776	27.4983
final_refit_pred_48h.csv	3,984	pred	0.0225	420.6625	8.6217	33.0443

The predictions folder also contains 8 pair_*_val_predictions.csv file(s) with row-level validation outputs for candidate analysis.

Demo-only evaluation files

These metrics were computed against any demo labels bundled with the repository. They are useful for quick checks, but they are not the official hidden-test challenge results.

file	overall_rmse	overall_mae	avg_county_rmse	n_rows	n_locations
demo_eval_24h.json	16.6618	4.9229	5.0823	1,992	83
demo_eval_48h.json	22.1493	6.0830	6.4129	3,984	83
demo_eval_final_refit_24h.json	28.4401	7.4314	7.9278	1,992	83
demo_eval_final_refit_48h.json	34.0336	8.2953	9.2121	3,984	83

Reusable model artifacts

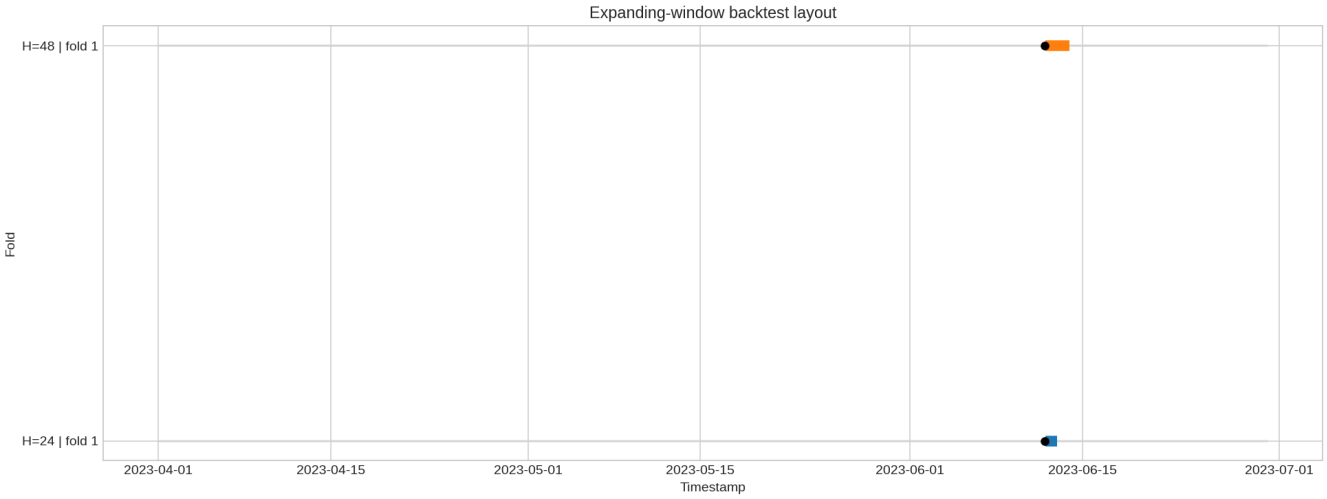
horizon	fold_id	pair_tag	classifier_name	regressor_name	two_stage_model_path	stage1_model_path	stage2_model_path
24	1	h24__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	pair_h24__clf_xgbc01__reg_xgbr01_f01_two_stage.joblib	pair_h24__clf_xgbc01__reg_xgbr01_f01_stage1_classifier.joblib	pair_h24__clf_xgbc01__reg_xgbr01_f01_stage2_regressor.joblib

horizon	fold_id	pair_tag	classifier_name	regressor_name	two_stage_model_path	stage1_model_path	stage2_model_path
48	1	h48__clf_xgbc01__reg_xgbr01	xgb_classifier	xgb_regressor	pair_h48__clf_xgbc01__reg_xgbr01_f01_two_stage.joblib	pair_h48__clf_xgbc01__reg_xgbr01_f01_stage1_classifier.joblib	pair_h48__clf_xgbc01__reg_xgbr01_f01_stage2_regressor.joblib

Visual appendix

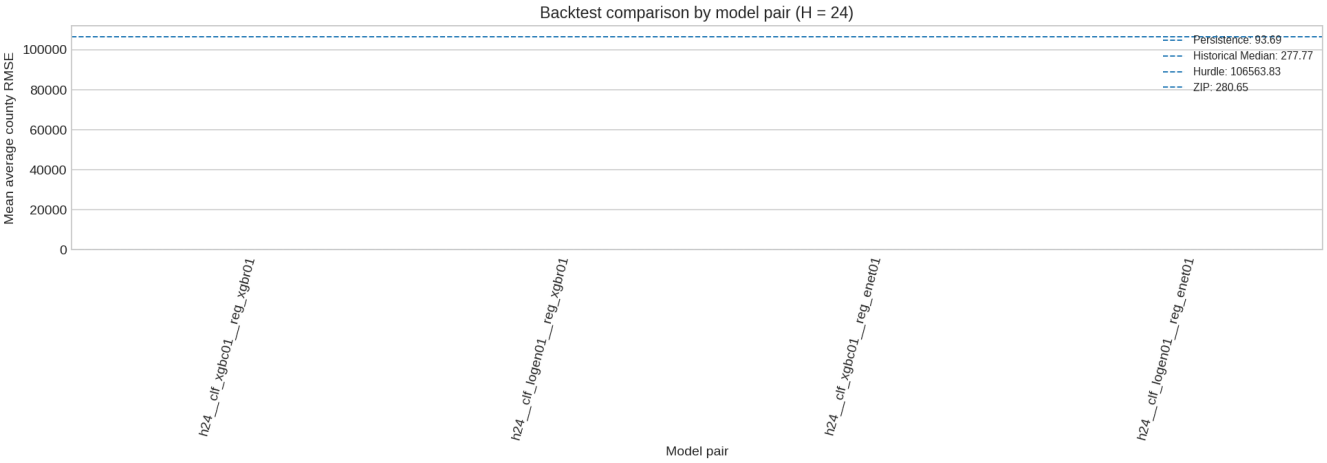
The following figures were read directly from the run's plots folder and inserted without GenAI interpretation.

Backtest Layout



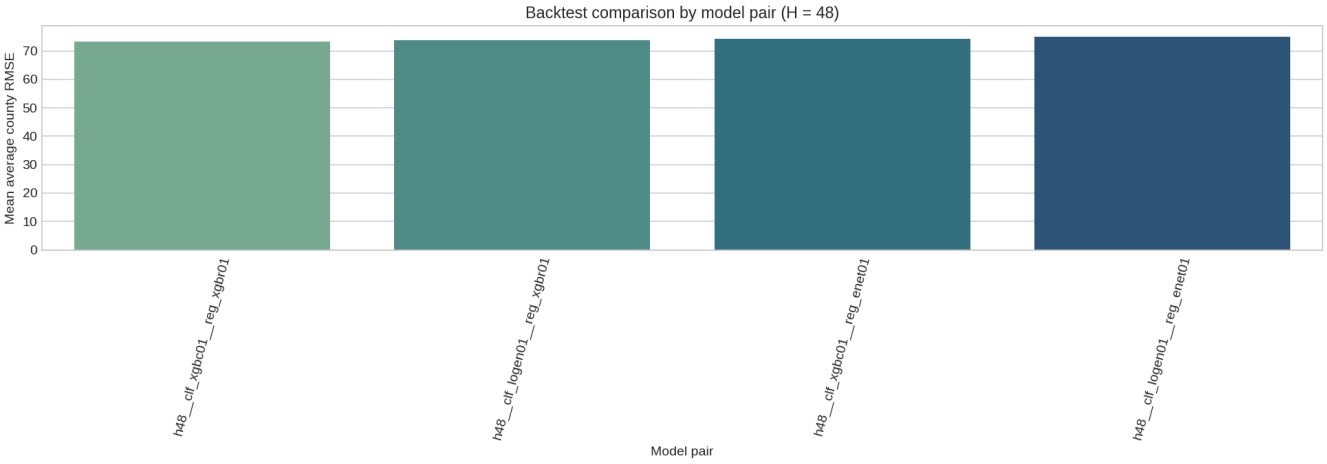
backtest_layout.png

Pair Comparison 24H



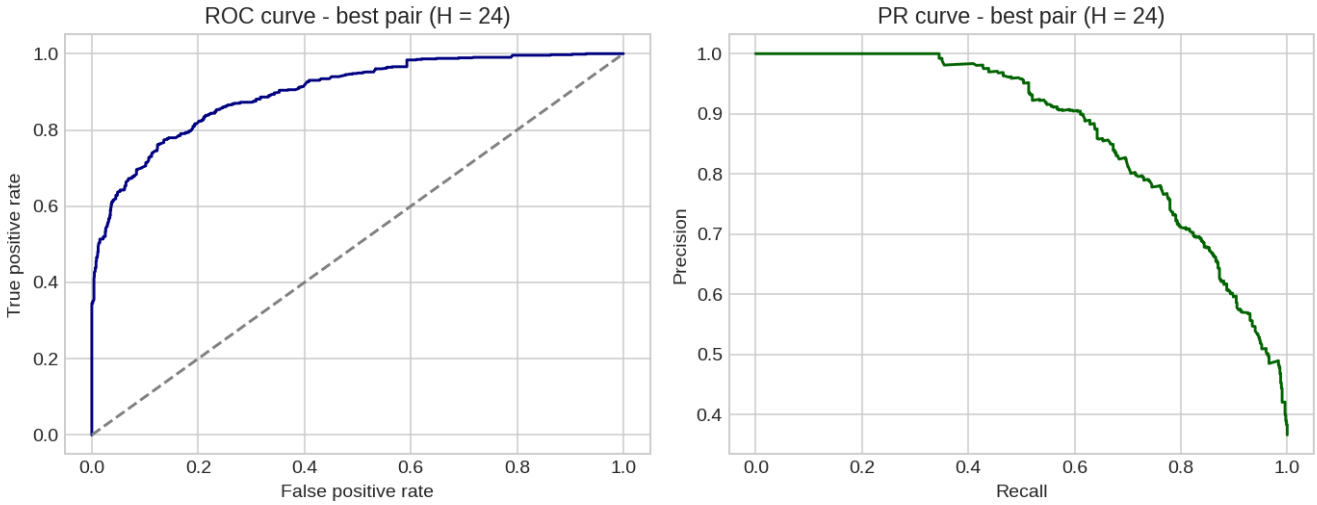
pair_comparison_h24.png

Pair Comparison 48H



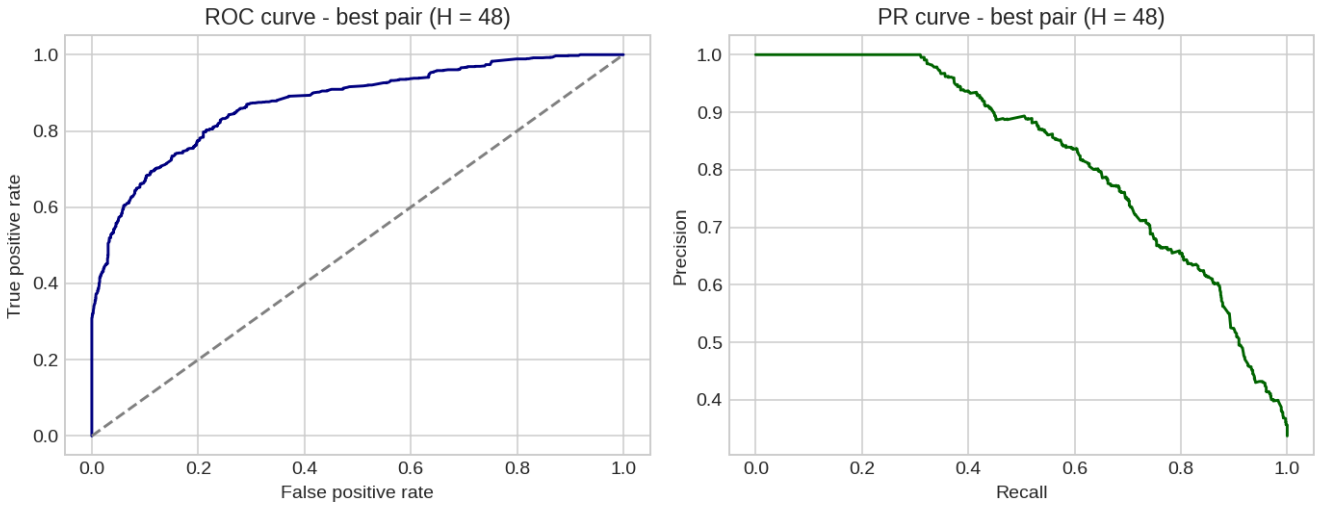
pair_comparison_h48.png

Best Pair Curves 24H



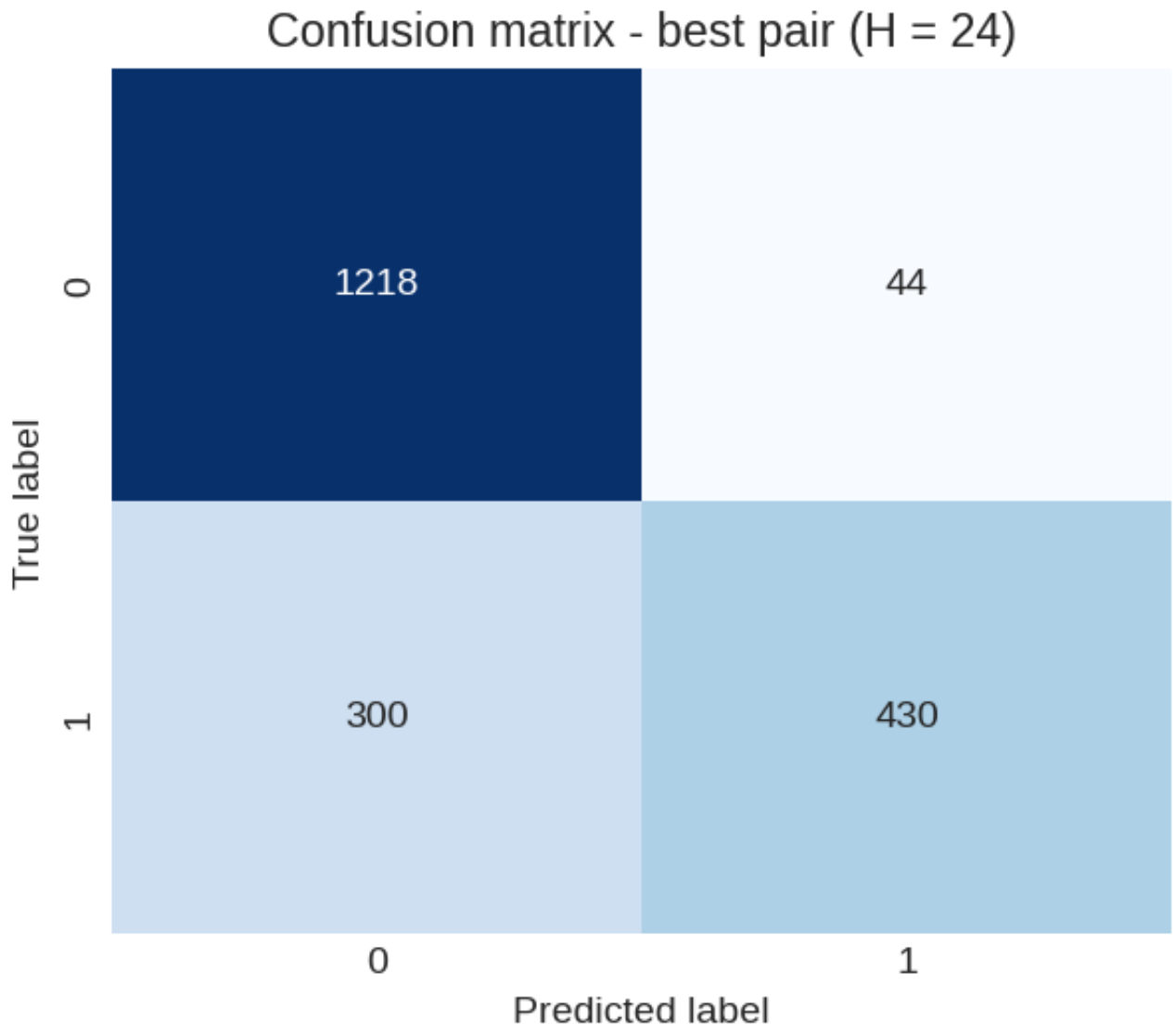
best_pair_curves_h24.png

Best Pair Curves 48H



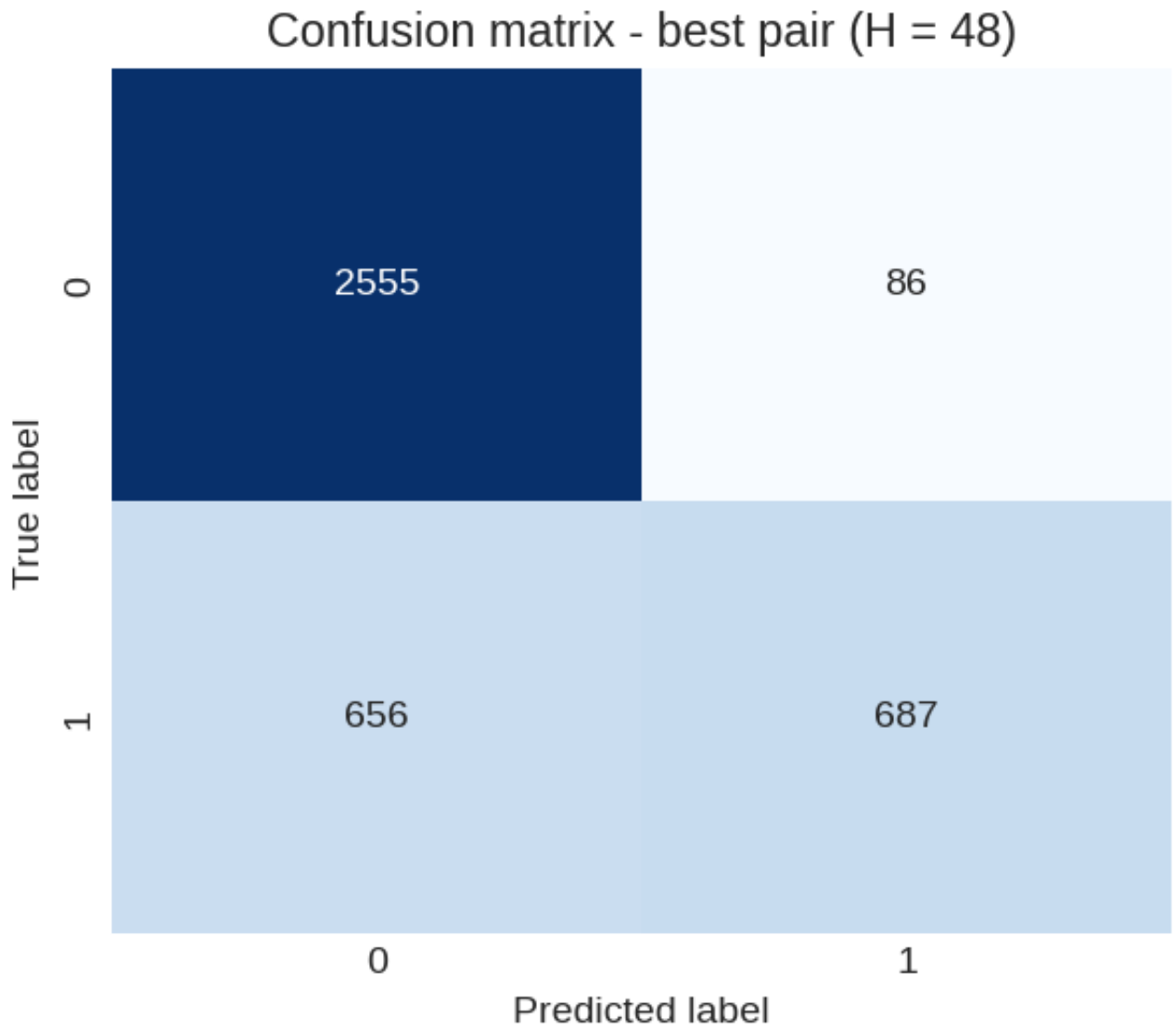
best_pair_curves_h48.png

Best Pair Confusion 24H



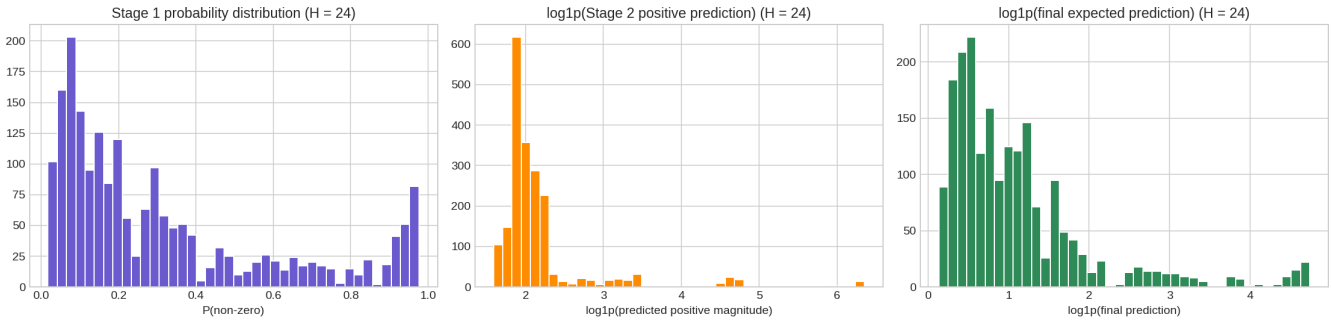
best_pair_confusion_h24.png

Best Pair Confusion 48H



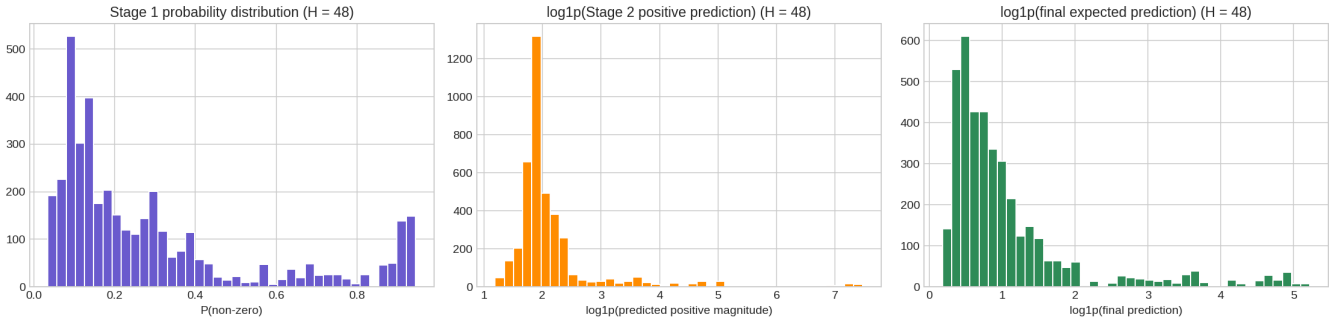
best_pair_confusion_h48.png

Prediction Distributions 24H



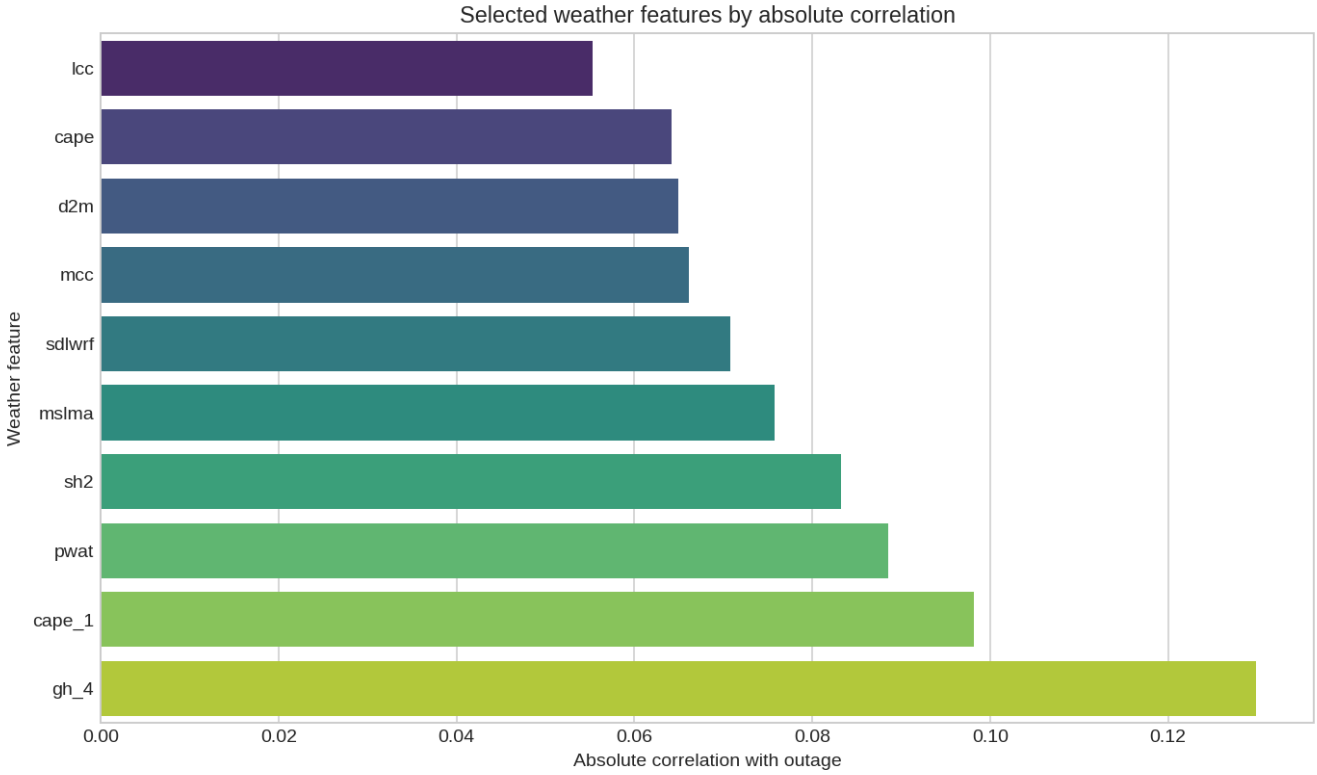
prediction_distributions_h24.png

Prediction Distributions 48H



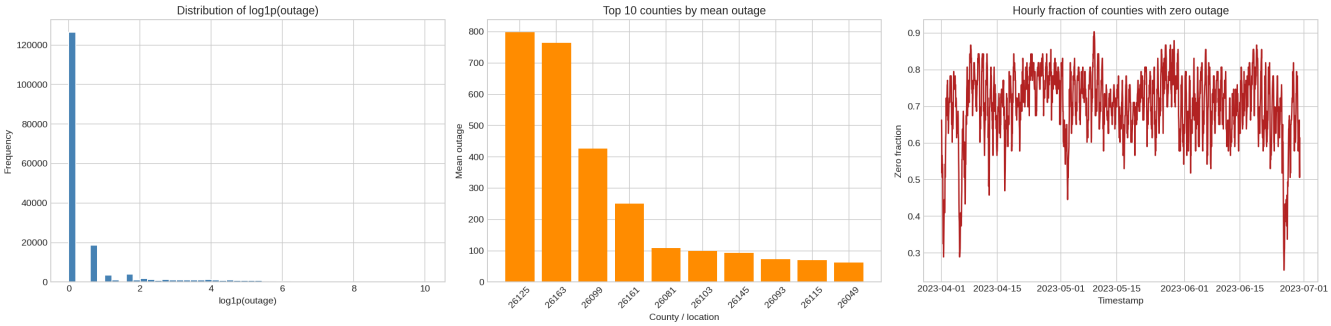
prediction_distributions_h48.png

Selected Weather Features



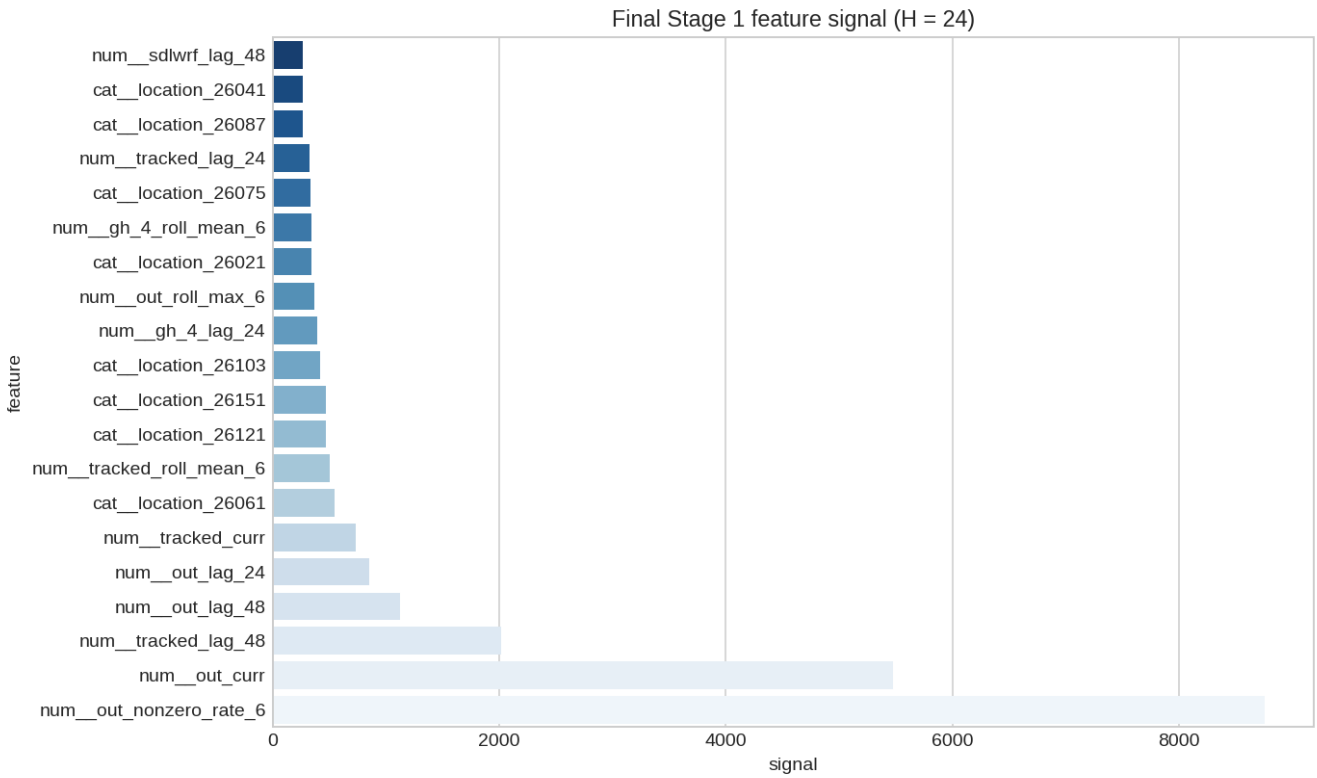
selected_weather_features.png

Eda Overview



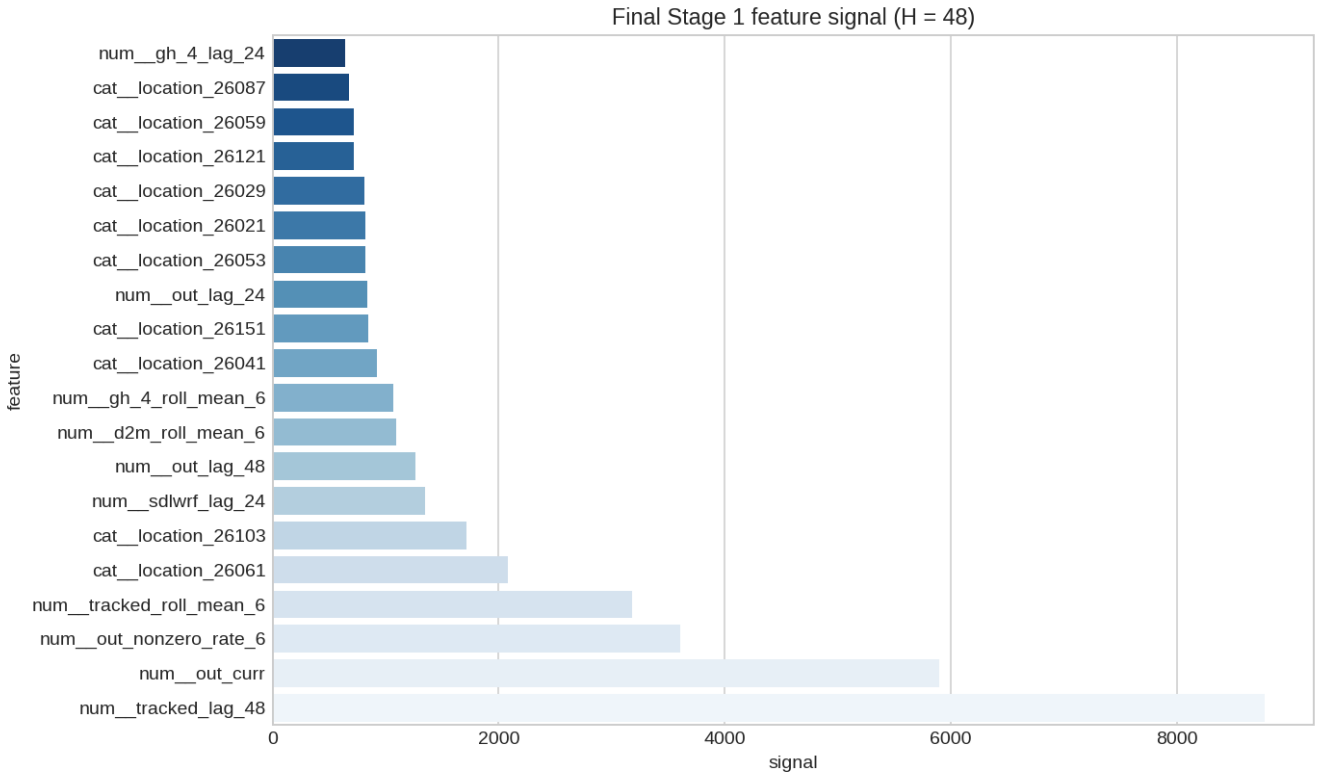
eda_overview.png

Final Stage1 Feature Signal 24H



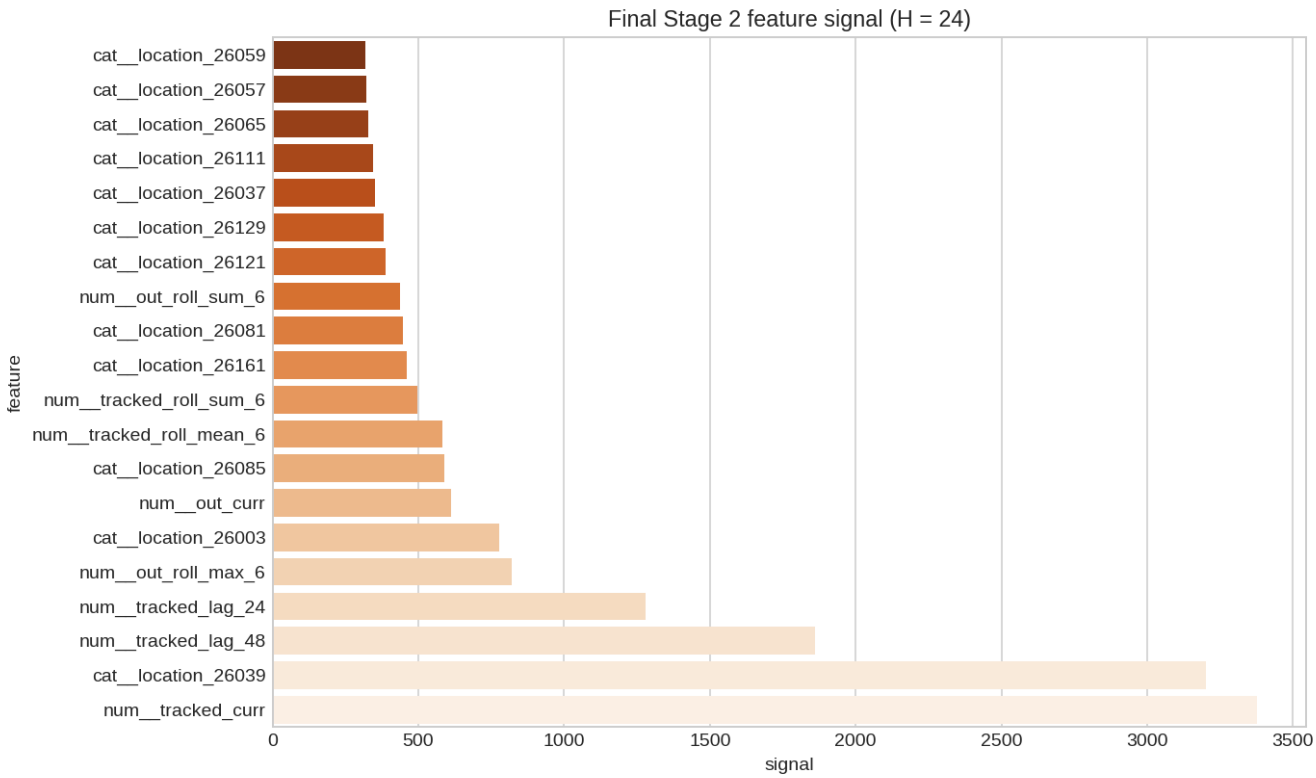
final_stage1_feature_signal_h24.png

Final Stage1 Feature Signal 48H



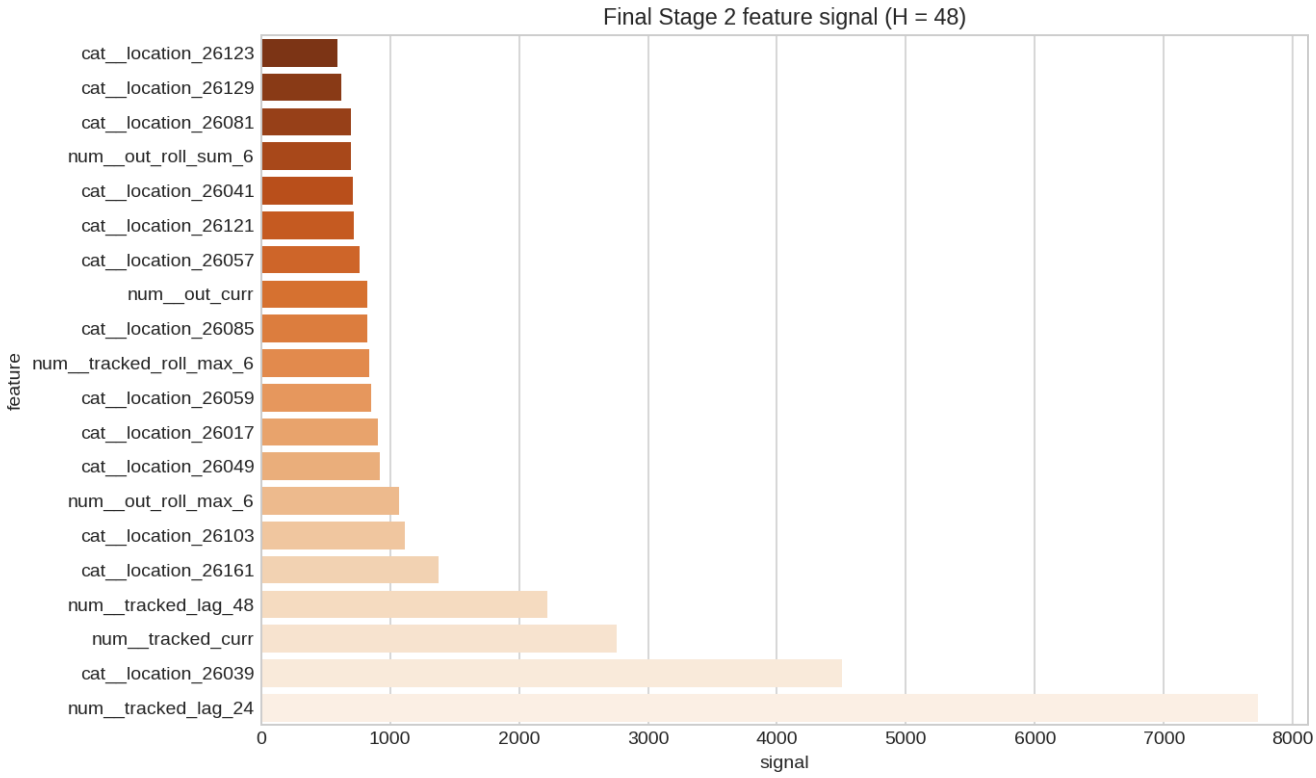
final_stage1_feature_signal_h48.png

Final Stage2 Feature Signal 24H



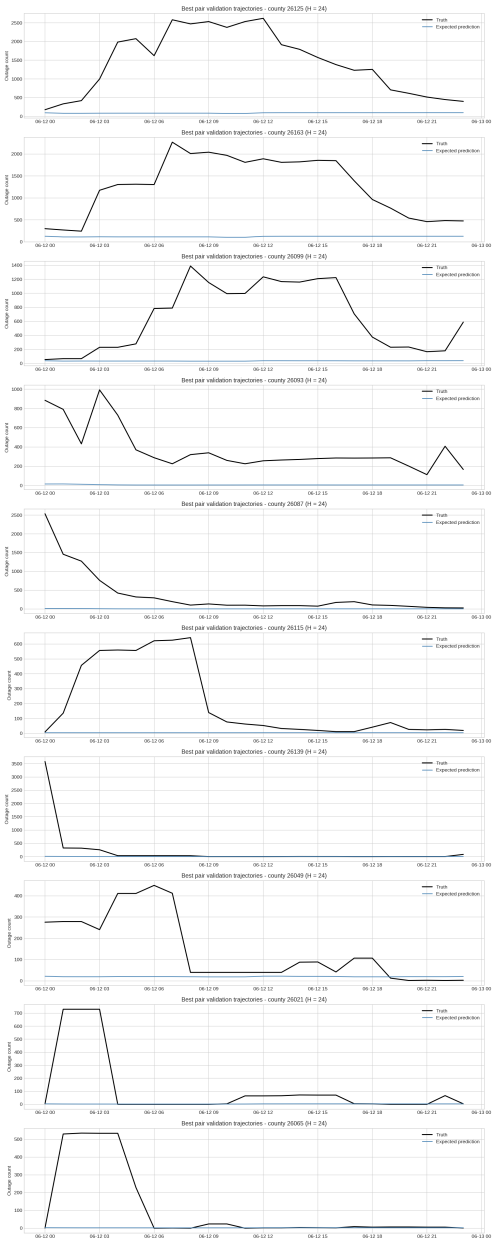
final_stage2_feature_signal_h24.png

Final Stage2 Feature Signal 48H



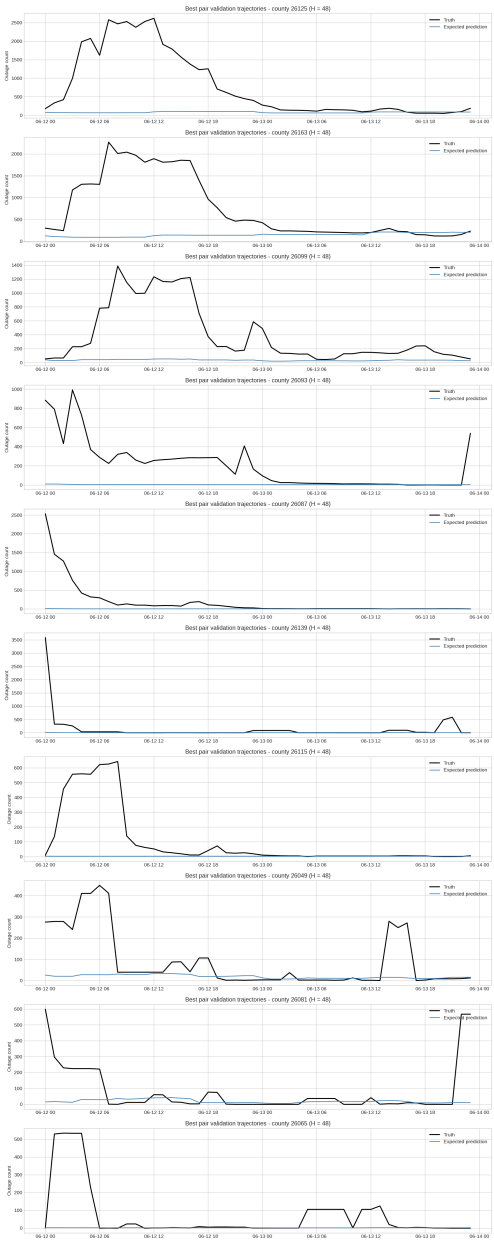
final_stage2_feature_signal_h48.png

Best Pair Validation Trajectories 24H



best_pair_validation_trajectories_h24.png

Best Pair Validation Trajectories 48H



best_pair_validation_trajectories_h48.png

Folder guide

- tables/: aggregated summaries, rankings, split definitions, and run-level CSV records.
- predictions/: exported submission-style predictions and detailed candidate validation predictions.
- models/: reusable .joblib artifacts plus metadata JSON files for submission and refit bundles.
- metrics/: fold-level metrics, training histories, and optional demo evaluation JSON files.
- plots/: visual diagnostics such as split layouts, model comparisons, confusion matrices, and feature-signal plots.
- checkpoints/: runtime resume artifacts used during notebook execution. This folder may be intentionally absent from GitHub uploads.